Description of data processing

2022-12-14

Contents

1 Introduction

The aim of this documentation is to illustrate how the data pre-processing is done for the data-set of the HP407 group project.

To replicate this study, you will need to install R, which is available at https://r-project.org/. For this preprocessing workflow, the current version of R on my computer is as follows:

```
R.Version()$version.string
```

```
## [1] "R version 4.2.2 (2022-10-31)"
```

To proceed, you must install and load the required packages, which is as follows:

```
# installation
install.packages("openxlsx")
install.packages("naniar")

# loading
library(openxlsx)
library(naniar)
```

2 Loading the dataset

Use the following code to load the dataset:

```
# load the dataset
dataset <- read.xlsx("Dataset.xlsx")

# replace the string value "-" with NA
dataset[dataset=="-"]  <- NA

# remove the rows all data item is NA
dataset <- dataset[rowSums(is.na(dataset)) != ncol(dataset), ]

# remove the columns all data item is NA
dataset <- dataset[,colSums(is.na(dataset))<nrow(dataset)]
```

When loading the dataset, R will alter some of the row names in case of incompatibility. After loading the dataset, we will first have a look at its size, with the following code:

```
# show the size of the dataset
dim(dataset)
```

```
## [1]  733 3705
```

It is clear that this is a very large dataset with 733 rows and 3705 columns. This means the dataset contains 733 studies. We can first have a look at the first five rows of the dataset, with the following code to know what the columns might be about. The output of the code is omitted, as they are too long, due to the very large number of columns.

```
# show first five rows
head(dataset)
```

The 3705 columns are not all useful to us, except for the first several columns containing basic information. Because it will be very difficult for us to read all the columns with either Excel or R, we can first output all these names to a seperate file with the following code:

```
sink("colnames.txt")
colnames(dataset)
sink()
```

3 Screening by intervention type

Here we aim to filter the studies by the intervention type. First we can have a look at the column names related to "Intervention":

```
df_cols <- colnames(dataset)
df_cols[grep('Intervention', df_cols)]
```

```
##  [1] "Intervention"
##  [2] "Intervention.types"
##  [3] "Intervention.subgroups"
##  [4] "Intervention.Brief.description.of.intervention"
##  [5] "Intervention.Brief.description.ofi.ntervention"
##  [6] "Intervention.Brief.intervention.description"
##  [7] "Intervention.Brief.introduction.to.the.intervention"
##  [8] "Intervention.Intervention.details"
##  [9] "Intervention.Intervention.details.1"
## [10] "Intervention.Intervention.details.(duration.and.intensity.of.the.intervention,.dosage.of.drugs
## [11] "Intervention.intervention.type"
## [12] "Intervention.Intervention.type.1"
## [13] "Intervention.Interv..Type:"
```

```
## [14] "Intervention.Interv..type.[SELECT:.Pharmaceutical;.Traditional.Chinese.Medicine;.Other.trad..me
## [15] "Intervention.Interv..type.[SELECT:.Pharmaceutical;.Traditional.Chinese.Medicine;.Other.trad..me
## [16] "Intervention.Name.of.intervention"
## [17] "Zarit.Burden.Interview.(ZBI).Post.Intervention.Change.mean"
## [18] "Zarit.Burden.Interview.(ZBI).Post.Intervention.Change.N"
## [19] "Zarit.Burden.Interview.(ZBI).Post.Intervention.Change.SD"
```

3.1 Intervention types

Among these variables, "Intervention types" seems to provide most relevant information for fast screening:

```
knitr::kable(table(dataset$'Intervention.types'))
```

Var1	Freq
Comprehensive care interventions	7
Electrical brain stimulation	14
Multicomponent interventions	28
Music interventions	8
No intervention	251
Nutrition	4
Other interventions	2
Pharmaceutical	130
Physical exercise	17
Structured therapeutic psychosocial interventions	38
Supplements	44
Support for carers	14
Supportive psychosocial interventions	8
Traditional Chinese Medicine	156
Training and education for carers	6
Treatment of co-morbidities or additional risks	4
Unstructured therapeutic interventions	2

Thus, we will select 'Support for carers' and 'Training and education for carers'. But in case of any missingness, we also explore the studies under the intervention types of 'Structured therapeutic psychosocial interventions', 'Comprehensive care interventions', 'Supportive psychosocial interventions', 'Treatment of co-morbidities or additional risks', 'Unstructured therapeutic interventions' and 'Multicomponent interventions', but there is none related to support or education for caregivers.

```
df_studies <- dataset[dataset$'Intervention.types' %in% c(
    "Training and education for carers",
    "Support for carers",
    "Comprehensive care interventions"), ]
dim(df_studies)
```

```
## [1]   27 3705
```

Thus, we reduce the scope of the analysis from 733 studies to only 27 of them. Then we can have a look at the further details of the intervention.

```
knitr::kable(table(df_studies$'Intervention.subgroups'))
```

Var1	Freq
Community-based interventions	1
Group support and therapy	7

Var1	Freq
Individual-based interventions	6
Information, education, and training to support carers	7
Training and education for professional carers	2
Training and education for unpaid carers	4

Thus, the following studies are selected:

```
# df_studies
selected_studies <- unique(df_studies$Study.Identifier)
rownames(df_studies) <- NULL
knitr::kable(df_studies[c('Study.Identifier', 'Intervention.types', 'Intervention.subgroups', 'Country')
```

Study.Identifier	Intervention.types	Intervention.subgroups	Country
Amit 2008	Training and education for carers	Training and education for unpaid carers	India
Pahlavanzadeh 2010	Support for carers	Information, education, and training to support carers	Iran
Wang 2010b	Comprehensive care interventions	Individual-based interventions	China
ZHAO 2010	Comprehensive care interventions	Individual-based interventions	China
TAN 2010	Training and education for carers	Training and education for unpaid carers	China
SUN 2010	Comprehensive care interventions	Individual-based interventions	China
Guerra 2011	Training and education for carers	Training and education for unpaid carers	Peru
Wang 2012e	Support for carers	Group support and therapy	China
SU 2012	Comprehensive care interventions	Individual-based interventions	China
JIANG 2012	Comprehensive care interventions	Individual-based interventions	China
Aboulafia-Brakha 2014	Support for carers	Group support and therapy	Brazil
Aboulafia-Brakha 2014	Support for carers	Information, education, and training to support carers	Brazil
Wang 2014a	Comprehensive care interventions	Individual-based interventions	China
Mercedeh 2014	Support for carers	Group support and therapy	Iran
Kamkhagi 2015	Support for carers	Information, education, and training to support carers	Brazil
Kamkhagi 2015	Support for carers	Group support and therapy	Brazil
Söylemez 2016	Support for carers	Information, education, and training to support carers	Turkey
Yang 2017a	Comprehensive care interventions	Community-based interventions	China
Wang 2017e	Training and education for carers	Training and education for professional carers	China
Wang 2017c	Training and education for carers	Training and education for professional carers	China
Shata 2017	Support for carers	Group support and therapy	Egypt

Study.Identifier	Intervention.types	Intervention.subgroups	Country
Salamizadeh 2017	Support for carers	Information, education, and training to support carers	Iran
Mahdavi 2017	Support for carers	Group support and therapy	Iran
Mahdavi 2017	Support for carers	Group support and therapy	Iran
Lök 2017	Support for carers	Information, education, and training to support carers	Turkey
Lin 2017	Training and education for carers	Training and education for unpaid carers	China
Orawan 2018	Support for carers	Information, education, and training to support carers	Thailand

4 Screening by column names

However, it will be hard in term of human labour to look thourgh the 3000+ columns for data extraction, so I use a series of keywords to screen the relevant columns for meta-analysis.

4.1 Reserved columns

Before we start, we reserved the first 19 columns as they contain the most essential information about the studies from our perspective.

```
reserved_cols <- head(df_cols,93)
reserved_cols
```

```
##  [1] "Study.Identifier"
##  [2] "Year"
##  [3] "Intervention"
##  [4] "Intervention.types"
##  [5] "Intervention.subgroups"
##  [6] "Country"
##  [7] "Setting"
##  [8] "Types.of.disease_clean"
##  [9] "Type.of.disease.in.two.categories.(dementia.and.MCI)"
## [10] "Updated.Dementia.Severity"
## [11] "Is.this.a.study.focusing.on.carers.of.people.living.with.dementia.or.mild.cognitive.impairment
## [12] "Primary.Cognition.Measure.standardized.z-score.change"
## [13] "Type"
## [14] "Baseline.Score.-.Primary.Cognition"
## [15] "Number.of.Participants.at.Baseline"
## [16] "Baseline.SD"
## [17] "Post.Treatment.Score"
## [18] "Post.Treatment.N"
## [19] "Post.Treatment.SD"
## [20] "Post.Treatment.Time.(Weeks.from.Start)"
## [21] "Post.Treatment.Time.(Months.from.Start)"
## [22] "Follow.up.N"
## [23] "Follow.Up.Score"
## [24] "Follow.up.SD"
## [25] "Follow.up.Time.(Weeks.since.end.of.study)"
## [26] "Follow.up.Time.(Months.since.end.of.study)"
## [27] "Domain.1"
## [28] "Domain.2"
```

```
## [29] "Domain.3"
## [30] "Domain.4"
## [31] "Domain.5"
## [32] "Overall"
## [33] "Baseline.Age.in.years.(mean).age.at.diagnosis"
## [34] "Baseline.Age.in.years.(range)"
## [35] "Baseline.Age.(years)"
## [36] "Baseline.APOE-e4.carriers.(percent)"
## [37] "Baseline.BMI.(mean)"
## [38] "Baseline.Clinical.dementia.rating.sum.of.boxes.(mean)"
## [39] "Baseline.Educational.level.<5.yrs.(%)"
## [40] "Baseline.Educational.level.(basic.education,.percent)"
## [41] "Baseline.Educational.level:.Elementary.school.or.lower.(percent)"
## [42] "Baseline.Educational.level.(had.formal.education)"
## [43] "Baseline.Educational.level:.High.school.(percent)"
## [44] "Baseline.Educational.level.(illiterate,.percent))"
## [45] "Baseline.Educational.level.(illiterate/primary.school/middle.school/college.degree.or.above).(
## [46] "Baseline.Educational.level.(indicate.how.this.was.measured)"
## [47] "Baseline.Educational.level.(indicate.how.this.was.measured):.years.of.schooling"
## [48] "Baseline.Educational.level.(longer.than.middle.school)"
## [49] "Baseline.Educational.level.(mean)"
## [50] "Baseline.Educational.level:.middle.school.or.above.(percent)"
## [51] "Baseline.Educational.level.(primary.school)"
## [52] "Baseline.Educational.level.(years)"
## [53] "Baseline.Educational.level.(years,.mean)"
## [54] "Baseline.Educational.level.(Years.of.schooling)"
## [55] "Baseline.Education,.College.(n)"
## [56] "Baseline.Education,.Diploma.(n)"
## [57] "Baseline.Education,.illiterate.(n)"
## [58] "Baseline.Education,.Lettered.(n)"
## [59] "Baseline.Education.level:.College.(percent)"
## [60] "Baseline.Education.level:.Middle.school.(%)"
## [61] "Baseline.Education.level:.no.school.(percent)"
## [62] "Baseline.Education.level:.primary.school.(percent)"
## [63] "Baseline.Education,.Middle.school.(n)"
## [64] "Baseline.Female.(percent)"
## [65] "Baseline.For.studies.in.carers.only:.Caregiver.age.in.years.(mean)"
## [66] "Baseline.For.studies.in.carers.only:.Caregiver.female.(percent)"
## [67] "Baseline.For.studies.in.carers.only:.Full-time.working.(percent)"
## [68] "Baseline.For.studies.in.carers.only:.Part-time.working.(percent)"
## [69] "Baseline.Mild.severity.of.dementia.(percent)"
## [70] "Baseline.MMSE.score.(mean)"
## [71] "Baseline.MMSE.score.(means)"
## [72] "Baseline.MoCA.score.(mean)"
## [73] "Baseline.Moderate.severity.of.dementia.(percent)"
## [74] "Baseline.NPI.score.(mean)"
## [75] "Baseline.Number.of.co-morbidities.(mean)"
## [76] "Baseline.Rural.setting.(percent)"
## [77] "Baseline.SKT.total.score.(mean)"
## [78] "Baseline.Urban.setting.(percent)"
## [79] "Baseline.VaDAS-cog.(mean)"
## [80] "Baseline.Years.duration.of.illness.(mean)"
## [81] "Intervention.Brief.description.of.intervention"
## [82] "Intervention.Brief.description.ofi.ntervention"
```

```
## [83] "Intervention.Brief.intervention.description"
## [84] "Intervention.Brief.introduction.to.the.intervention"
## [85] "Intervention.Intervention.details"
## [86] "Intervention.Intervention.details.1"
## [87] "Intervention.Intervention.details.(duration.and.intensity.of.the.intervention,.dosage.of.drugs
## [88] "Intervention.intervention.type"
## [89] "Intervention.Intervention.type.1"
## [90] "Intervention.Interv..Type:"
## [91] "Intervention.Interv..type.[SELECT:.Pharmaceutical;.Traditional.Chinese.Medicine;.Other.trad..m
## [92] "Intervention.Interv..type.[SELECT:.Pharmaceutical;.Traditional.Chinese.Medicine;.Other.trad..m
## [93] "Intervention.Name.of.intervention"
```

4.2 Keyword screening

We note that there are quite a lot of depression/caregiver burden related metrics also in the columns. So we output the colnames to a seperate file named colnames.txt for manual screening of useful columns. According to our preliminary screening, we notice a number of metrics to measure caregiver burden.

4.2.1 Direct measure of care giver burdern

First we screen the column names with the keywords for direct measurement of caregiver burden

- **Hinton 2020** and **Uyar 2019:** Zarit Burden Interview (ZBI)
- **Govindakumari 2020:** Care giver burden, unspecified

```
caregiver_burden_keys <- unique(c(
    grep("Burden", df_cols, ignore.case=TRUE),
    grep("ZBI", df_cols, ignore.case=TRUE),
    grep("Zarit", df_cols, ignore.case=TRUE)
    ))
caregiver_burden_keys <- df_cols[caregiver_burden_keys]

# show the number of selected columns
length(caregiver_burden_keys)
```

```
## [1] 91
```

There are a total of 91 columns selected. We will look at all the columns selected but for brevity of this article, we omit the output of the following code.

```
caregiver_burden_keys
```

4.2.2 Neuropsychiatric symptom scales

Similar procedures are used to screen the colunms related to neuropsychiatric symptoms of caregivers.

Neuropsychiatric symptoms of caregivers:

- **Hinton 2020:** Depression/anxiety symptoms, measured with the Patient Health Questionnaire-4 (PHQ-4)
- **Uyar 2019:** Depression symptoms, measured with Beck Depression Inventory (BDI)
- **Uyar 2019:** Anxiety symptoms, measured with Beck Anxiety Inventory (BAI)
- **Uyar 2019:** Neuropsychiatric symptoms, measured with Neuropsychiatric Inventory-Distress (NPI-D)
- **Ghaffari 2019:** Neuropsychiatric symptoms, measured with General Health Questionnaire (GHQ-28)
- **Zarepour 2020:** Anxiety symtoms, measured with the Spielberger questionnaires
- **Pan 2019:** Depression symptoms, measured with 10-item CES-D
- **Uyar 2019:** Quality of life, measured with Quality of Life Scale SF36 (SF36 mental health)

Neuropsychiatric symptoms of patients:

- **Chen 2020** and **Pan 2019:** Mini Mental State Examinination (MMSE)
- **Uyar 2019:** Neuropsychiatric symptoms, measured with Neuropsychiatric Inventory–Severity (NPI-S)

```
# keywords for neuropsychiatric symptoms:
mental_health_keys <- unique(c(

    # abbreviations
    grep("PHQ", df_cols, ignore.case=TRUE),
    grep("SF36", df_cols, ignore.case=TRUE),
    grep("NPI", df_cols, ignore.case=TRUE),
    grep("GHQ", df_cols, ignore.case=TRUE),
    grep("PHQ", df_cols, ignore.case=TRUE),
    grep("BDI", df_cols, ignore.case=TRUE),
    grep("BAI", df_cols, ignore.case=TRUE),
    grep("MMSE", df_cols, ignore.case=TRUE),

    # words and phrases
    grep("neuropsychiatric", df_cols, ignore.case=TRUE),
    grep("depression", df_cols, ignore.case=TRUE),
    grep("anxiety", df_cols, ignore.case=TRUE)
    ))

# select mental health keys and drop columns and rows with only NAs
mental_health_keys <- df_cols[mental_health_keys]

# show the number of selected columns
length(mental_health_keys)
```

```
## [1] 705
```

4.2.3 Behavioural metrics

Behavioural metrics of caregivers:

- **Uyar 2019:** Quality of life, measured with Quality of Life Scale SF36 (SF36 physical health)
- **Pan 2019:** Positive/Negative coping, measured with the Simplified Coping Scale
- **Pan 2019:** Mutuality, measured with 15-item Mutuality Scale

Behavioural metrics of patients:

- **Uyar 2019:** Quality of life, measured with Quality of life in Alzheimer's Disease (QoL-AD)
- **Chen 2020:** Barthel activities of daily living scale (BADL)
- **Chen 2020:** Behavioral pathology assessment scale of Alzheimer's disease (BEHAVE-AD)
- **Govindakumari 2020:** Quality of life, unspecified
- **Pan 2019:** Quality of life, measured with the 14-item Activities of Daily Living scale (ADL)

```
# keywords for behavioural metrics:
behvaioural_keys <- unique(c(

    # abbreviations
    grep("SF36", df_cols, ignore.case=TRUE),
    grep("QoL", df_cols, ignore.case=TRUE),
    grep("BADL", df_cols, ignore.case=TRUE),
    grep("BEHAVE", df_cols, ignore.case=TRUE),
    grep("ADL", df_cols, ignore.case=TRUE),
```

```
    # words and phrases
    grep("Quality", df_cols, ignore.case=TRUE),
    grep("Coping", df_cols, ignore.case=TRUE),
    grep("Mutuality", df_cols, ignore.case=TRUE),
    grep("Daily", df_cols, ignore.case=TRUE),
    grep("Barthel", df_cols, ignore.case=TRUE),
    grep("Behaviour", df_cols, ignore.case=TRUE),
    grep("Behavior", df_cols, ignore.case=TRUE)
    ))

# select behavioural keys and drop columns and rows with only NAs
behvaioural_keys <- df_cols[behvaioural_keys]
# behvaioural_keys[-grepl('sleep', behvaioural_keys)]

# show the number of selected columns
length(behvaioural_keys)
```

```
## [1] 362
```

4.3 Subsetting the dataset

We first simplify the selected studies from a data frame to a list of study identifier.

```
selected_studies_list <- unique(df_studies$Study.Identifier)
length(selected_studies_list)
```

```
## [1] 24
```

Then we can subset the dataset according to our selection of studies and columns. For example, for the study that directly measures caregiver burden, we can use the following code:

```
# subset by caregiver burden-related columns
df_caregiver_burden <-
  dataset[which(rowSums(is.na(dataset[caregiver_burden_keys]))
              != ncol(dataset[caregiver_burden_keys])), ]
# subset by selected studies
df_caregiver_burden <-
  df_caregiver_burden[df_caregiver_burden$'Study.Identifier'
                      %in% as.list(selected_studies_list), ]

# drop rows and columns with only NA
df_caregiver_burden <-
  df_caregiver_burden[rowSums(is.na(df_caregiver_burden))
                      != ncol(df_caregiver_burden), ]
df_caregiver_burden <-
  df_caregiver_burden[,colSums(is.na(df_caregiver_burden))
                      < nrow(df_caregiver_burden)]

# show the size of simplified dataset
dim(df_caregiver_burden)
```

```
## [1]  20 152
```

Thus, the dataset is significantly reduced, which we can manually curate for a meta analysis-ready dataset. Similary, we can generate the dataset for caregiver mental health and caregiver quality of life (behaviour):

```r
# subset for caregiver mental health
df_mental_health <-
  dataset[which(rowSums(is.na(dataset[mental_health_keys]))
              !=ncol(dataset[mental_health_keys])), ]
df_mental_health <-
  df_mental_health[df_mental_health$'Study.Identifier'
                  %in% as.list(selected_studies_list), ]
df_mental_health <-
  df_mental_health[rowSums(is.na(df_mental_health))
                  != ncol(df_mental_health), ]
df_mental_health <-
  df_mental_health[,colSums(is.na(df_mental_health))
                  < nrow(df_mental_health)]

# print dimensions of the dataset
dim(df_mental_health)
```

```
## [1]  18 145
```

```r
# subset for caregiver behaviour or quality of life
df_behavioural <-
  dataset[which(rowSums(is.na(dataset[behvaioural_keys]))
              !=ncol(dataset[behvaioural_keys])), ]
df_behavioural <-
  df_behavioural[df_behavioural$'Study.Identifier'
                %in% as.list(selected_studies_list), ]
df_behavioural <-
  df_behavioural[rowSums(is.na(df_behavioural))
                != ncol(df_behavioural), ]
df_behavioural <-
  df_behavioural[,colSums(is.na(df_behavioural))
                < nrow(df_behavioural)]

# print dimensions of the dataset
dim(df_behavioural)
```

```
## [1]  14 151
```

We further create a dataset with only the necessary columns to allow further manual clean-up of demographic and other information of each study.

```r
# subset by selected studies
df_selected <-
  dataset[dataset$'Study.Identifier' %in%
          as.list(selected_studies_list), ]
df_selected <- df_selected[reserved_cols]
```

5 Saving the filtered dataset

Eventually, we can write these subsets to a single Excel file for further curation, with the following code:

```r
# create the workbook
wb <- createWorkbook()

# add the worksheets
```

```
addWorksheet(wb, "info")
addWorksheet(wb, "caregiver_Burden")
addWorksheet(wb, "caregiver_MentalHealth")
addWorksheet(wb, "caregiver_Behaviour")

# write the workbook
writeData(wb, "info", df_selected)
writeData(wb, "caregiver_Burden", df_caregiver_burden)
writeData(wb, "caregiver_MentalHealth", df_mental_health)
writeData(wb, "caregiver_Behaviour", df_behavioural)

# save the workbook
saveWorkbook(wb, "Dataset_data_filtered_final.xlsx", overwrite = TRUE)
```